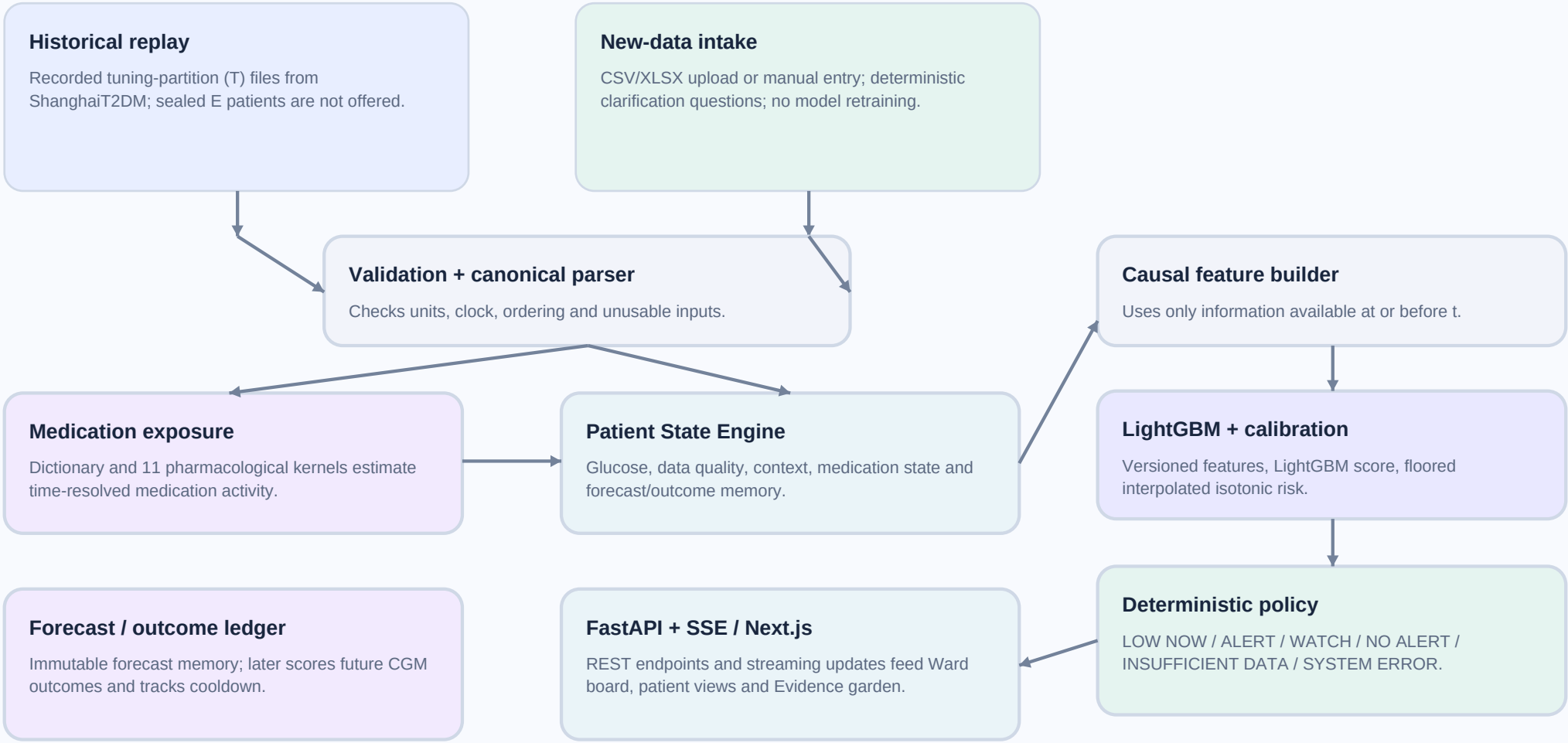
01 | End-to-end runtime

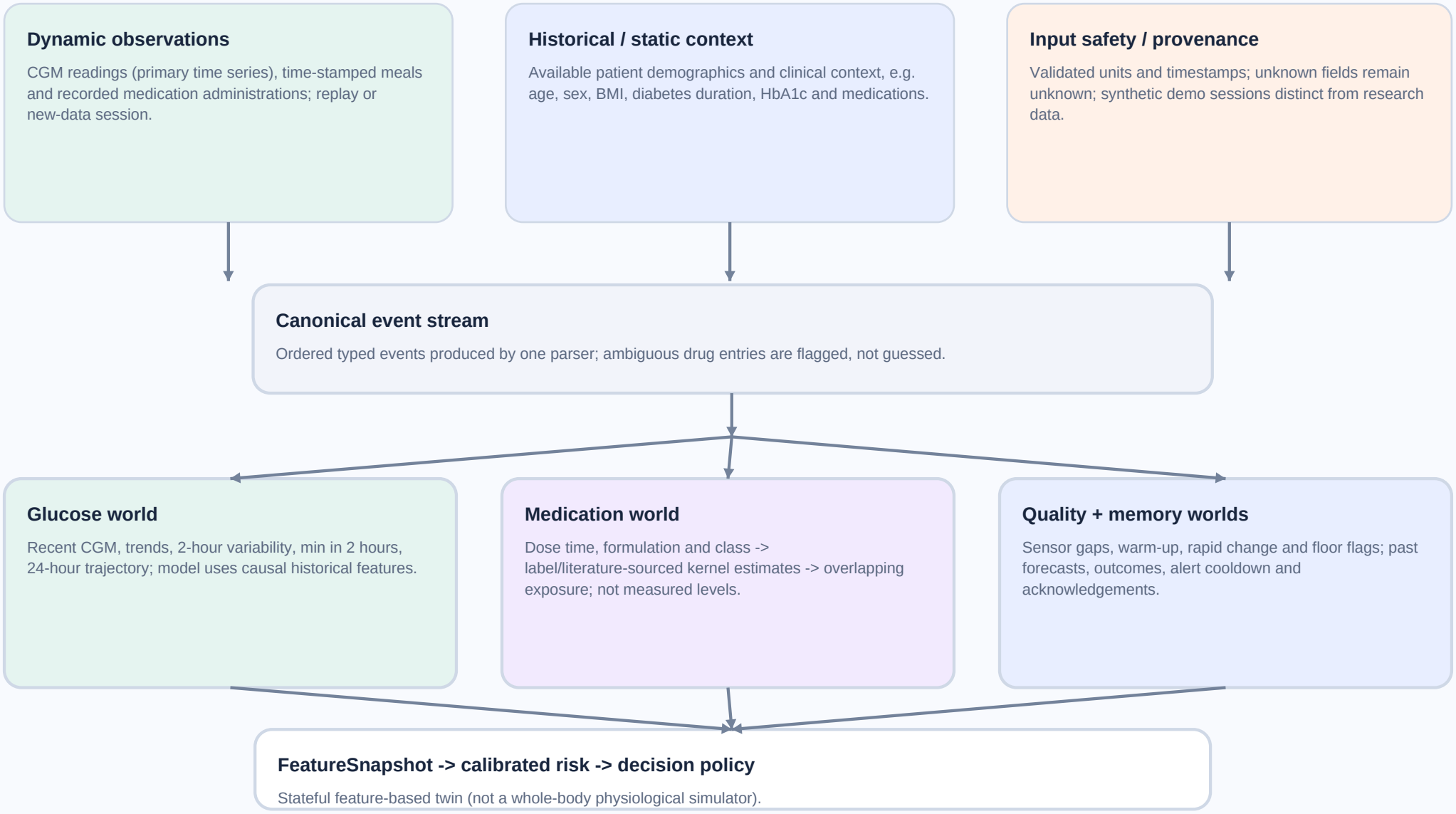
One inference path for recorded replay and new-data sessions - offline, CPU-based prototype



Clinical boundary: Human review and capillary glucose confirmation only. No dose recommendations or automated treatment. Research prototype; not clinically validated.

02 | Digital twin state & medication modelling

Patient state is continuously updated as each accepted observation arrives in time order



03 | Training, evaluation & evidence governance

Development evidence and sealed confirmatory evaluation are strictly separated

Official ShanghaiT2DM v5

Official dataset fetch; hashes verified against data/MANIFEST.json.

→

Patient-level split

T: 30 development patients; E: 70 sealed patients.

→

Development on T only

Nested patient-grouped CV, Optuna, calibrated model ladder.

→

Freeze candidate

M3 development candidate rebuilt twice with identical result hash.

→

Human gates

Clinical-pharmacist kernel review + operational-definition approval pending.

→

Confirmatory run

E only after verified freeze; not performed as of README export.

Model ladder

M0: deterministic CGM trend | M1: CGM features (38) | M2: + context/meal (72) | M3r: + raw medication timing (95) | M3: + exposure kernels (100).

Pre-registered endpoint

Sensitivity at 1 false ALERT / patient-day; episodes warned 15-60 min before CGM-detected onset. Primary comparison: M3 - M2 on sealed E-treated patients.

Development results on T - not confirmatory

At 1 FA/day, sensitivity: M0 62.6%; M1 85.7%; M2 87.5%; M3r 92.9%; M3 91.1%. M3 - M2 = +3.6 percentage points (95% CI -4.2 to +8.8); not a reliable demonstrated benefit. E remains sealed.

04 | Clinician workflow, interface & safety

A conceptual care-team review experience backed by running replay and new-data application

1. Observe / intake

Start historical replay or enter new data. Canonical parsing validates timestamps, CGM units and optional records.

2. Monitor the ward

Ward board sorts patient statuses and risk; latest CGM, trend, 60-minute probability and freshness visible.

3. Review patient

24-hour glucose/risk view separates observed LOW NOW from predicted next-60-min status.

4. Explain / inspect

Grouped TreeSHAP associations, Twin anatomy and estimated medication activity; not causal medical explanations.

5. Human confirmation

ALERT prompts review and capillary confirmation. No autonomous dosing, diagnoses or treatment.

6. Record outcomes

Ledger retains forecasts and later marks low/no low/pending/unknown when outcome window is available.

Constraints and system invariants

Missing or unreliable CGM yields INSUFFICIENT DATA, never reassuring NO ALERT. New uploads stay in memory; no retraining and no accuracy claim for new sessions. CGM-detected lows are not necessarily capillary-confirmed. Evidence is internal development only; confirmatory evaluation and pharmacist review remain pending.